

ROMIR MALIK

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[Portfolio](#); [Github](#); [Linkedin](#)

TECHNICAL SKILLS

- **Programming Languages:** Python (Pytorch, Pandas, NumPy, OpenCV, matplotlib), R, C++ (STL, PCL), SQL, Bash, MATLAB, CUDA
- **Data Engineering & AI:** Data Wrangling, ML Algorithms, Deep Learning, Text and Media Analysis, Transformers, Spatial Data and simulation modeling, Deep Reinforcement learning (DRL)
- **Robotics & Vision:** ROS/ROS2, Gazebo, Control Systems, Sensor Fusion, Navigation Stack, MoveIt, LeRobot

ACADEMIC QUALIFICATION

Utrecht University, MSc Applied Data Science

2025 - 2026

- **Coursework:** Data Wrangling, Text and Media Analysis, Transformers, Spatial Data Analysis and simulation modeling

Manipal University Jaipur, B.Tech in Data Science and Engineering

2020 - 2024

- **Coursework:** Multivariable Calculus, Differential Equations, stats, Data Structures and Algorithms

PROFESSIONAL EXPERIENCE

Machine Learning and Robotics Intern

Sept 2023 - May 2024

CSIR National Institute of Oceanography (NIO), Goa

- Engineered a Deep Learning-based object detection pipeline for Unmanned Underwater Vehicles (UUVs), reducing collision risks by 95% in simulated cluttered environments.
- Architected a monocular visual SLAM framework integrating ZoeDepth and ORB-SLAM3, achieving sub-5cm mapping accuracy for underwater 3D reconstruction.
- Optimized autonomous navigation logic using Artificial Potential Fields (APF) algorithms, decreasing path-planning latency by 30% during obstacle avoidance maneuvers.
- Deployed a high-fidelity Gazebo/ROS simulation environment to validate sensor fusion algorithms across 50+ test scenarios, accelerating the deployment cycle for physical research vessels.

Research Intern

May 2023 - Aug 2023

Indian Space Research Organization (ISRO), Space Application Centre

- Developed a high-performance software stack for PS-PL (Processing System-Programmable Logic) communication on Xilinx Zynq MPSoC, accelerating Deep Learning inference speeds by 4x compared to CPU execution.
- Implemented hardware-accelerated Real-Time Object Detection on FPGA fabric, achieving 60+ FPS throughput for low-latency onboard vision applications.
- Benchmarked MPSoC power-performance metrics, delivering a feasibility report adopted by senior SAC scientists for future satellite edge-computing architectures.

RESEARCH PUBLICATIONS

- Rope augmented path following and control of remotely operated underwater vehicle using vision for stilling basin surveillance

IEEE APSCON 2024 [\[link\]](#)

Jan 2024

PROJECTS

Real-Time Amazon Product Detection with Few-Shot Object Detection [\[link\]](#)

Oct 2024

[Tech Stack: Pytorch, YOLO, FLASK, FSOD, HTML, Tailwind CSS, JavaScript, RESTful API, Next.js]

- Developed a system to detect Amazon products in Prime Video with only 5-10 Images per product class using Few-Shot Object Detection.
- Created a Custom Dataset for Training using innovative Augmentation Strategies and enabled purchased links for detected products in movies and web series available on Prime Video.
- Accelerated inference latency by migrating the model architecture from PyTorch to ONNX Runtime, enabling real-time detection rates on standard hardware.

Optimized Deep Learning Inspired MRI-based Segmentation for Alzheimer's Detection [\[link\]](#)

May 2023

[Tech Stack: Tensorflow, MATLAB, Python]

- Conducted extensive research on current instance segmentation techniques used in bio medical field, ultimately compiling various effective methods and models to process 3D MRI slices
- Built and optimized custom Resnet50 using Tensorflow library on various large data sets including ADNI, trained the model to achieve IOU of 0.85 and val accuracy of 0.87 in 25 epochs.

Leadership and Extra Curricular

Head of External Affairs

Sept 2025 - Present

Utrecht Data Science Community

- Spearheaded strategic partnerships with 10+ industry stakeholders to bridge the academia-industry gap, resulting in a 25% increase in student-professional mentorship engagements.
- Orchestrating the end-to-end execution of large-scale symposiums and in-house days, managing cross-functional teams to boost attendee participation by 40% year-over-year.
- Established a robust networking pipeline connecting 500+ data science students with corporate recruiters, directly facilitating internship placements and knowledge transfer.

CERTIFICATIONS

- **Bloomberg Market Concepts** [\[link\]](#)
Bloomberg

Oct 2025

- **JP Morgan Quantitative Research Job Simulation** [\[link\]](#)
Forage

July 2025

- **R Programming** [\[link\]](#)
John Hopkins University

May 2025

- **Fast ai Deep Learning For Coders** [\[link\]](#)
Jeremy Howard

Jan 2024